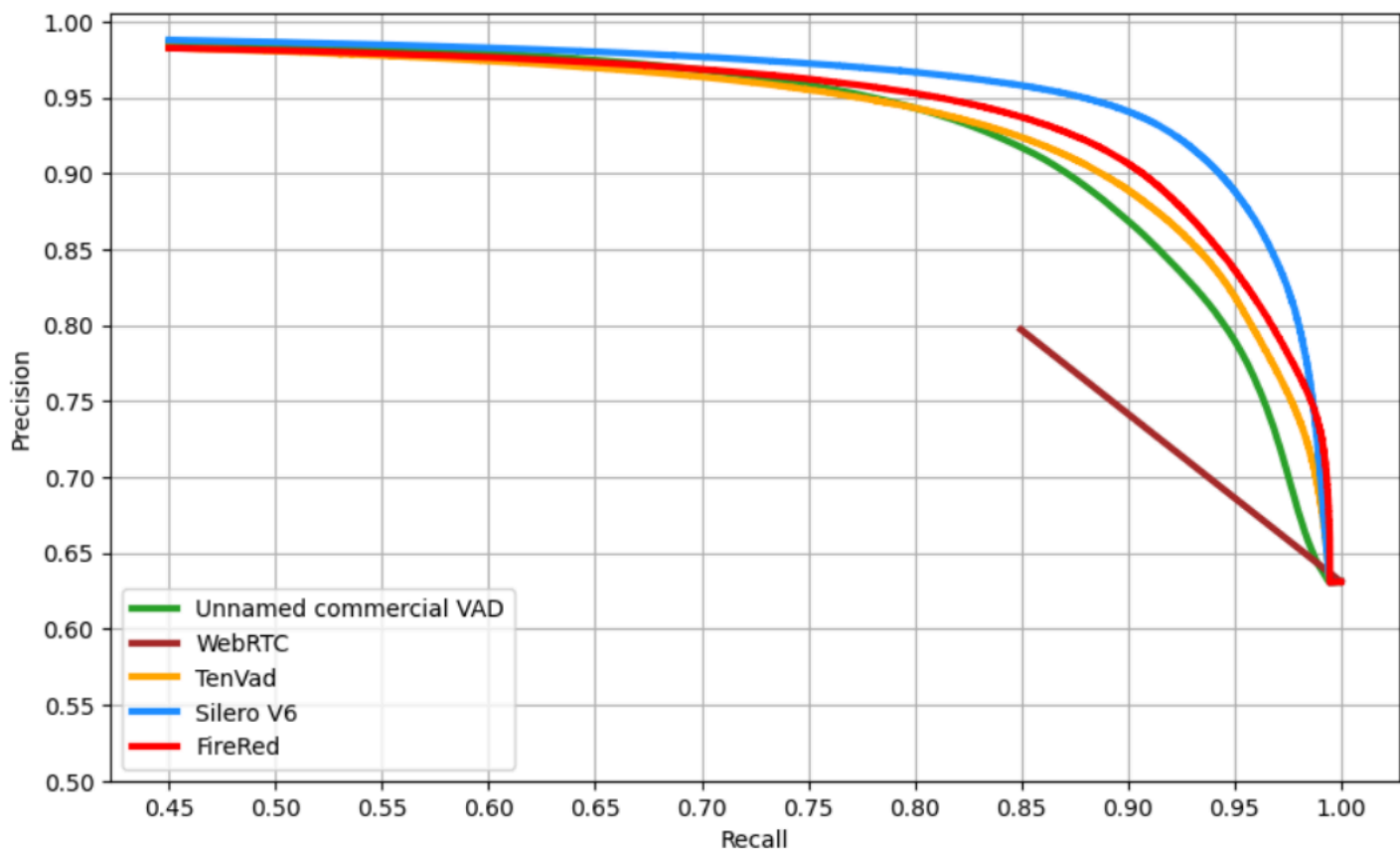




Silero VAD

Silero VAD - pre-trained enterprise-grade [Voice Activity Detector](#) (also see our [STT models](#)).

Comparison with other VAD models
Precision-Recall curve, Multi-Domain Validation



► Real Time Example

► Dependencies

Using pip: `pip install silero-vad`

```
from silero_vad import load_silero_vad, read_audio, get_speech_timestamps
model = load_silero_vad()
wav = read_audio('path_to_audio_file')
speech_timestamps = get_speech_timestamps(
    wav,
    model,
    return_seconds=True, # Return speech timestamps in seconds (default is
samples)
)
```

Using torch.hub:

```
import torch
torch.set_num_threads(1)

model, utils = torch.hub.load(repo_or_dir='snakers4/silero-vad',
model='silero_vad')
(get_speech_timestamps, _, read_audio, _, _) = utils

wav = read_audio('path_to_audio_file')
speech_timestamps = get_speech_timestamps(
    wav,
    model,
    return_seconds=True, # Return speech timestamps in seconds (default is
samples)
)
```

Key Features

- **Stellar accuracy**

Silero VAD has [excellent results](#) on speech detection tasks.

- **Fast**

One audio chunk (30+ ms) [takes](#) less than **1ms** to be processed on a single CPU thread. Using batching or GPU can also improve performance considerably. Under certain conditions ONNX may even run up to

4-5x faster.

- **Lightweight**

JIT model is around two megabytes in size.

- **General**

Silero VAD was trained on huge corpora that include over **6000** languages and it performs well on audios from different domains with various background noise and quality levels.

- **Flexible sampling rate**

Silero VAD [supports](#) **8000 Hz** and **16000 Hz** [sampling rates](#).

- **Highly Portable**

Silero VAD reaps benefits from the rich ecosystems built around **PyTorch** and **ONNX** running everywhere where these runtimes are available.

- **No Strings Attached**

Published under permissive license (MIT) Silero VAD has zero strings attached - no telemetry, no keys, no registration, no built-in expiration, no keys or vendor lock.

Typical Use Cases

- Voice activity detection for IOT / edge / mobile use cases
- Data cleaning and preparation, voice detection in general
- Telephony and call-center automation, voice bots
- Voice interfaces

Links

- [Examples and Dependencies](#)
- [Quality Metrics](#)
- [Performance Metrics](#)
- [Versions and Available Models](#)
- [Further reading](#)
- [FAQ](#)

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Please see our [wiki](#) for relevant information and [email](#) us directly.

```
@misc{Silero VAD,  
  author = {Silero Team},  
  title = {Silero VAD: pre-trained enterprise-grade Voice Activity Detector  
(VAD), Number Detector and Language Classifier},  
  year = {2024},  
  publisher = {GitHub},  
  journal = {GitHub repository},  
  howpublished = {\url{https://github.com/snakers4/silero-vad}},  
  commit = {insert_some_commit_here},  
  email = {hello@silero.ai}  
}
```

Examples and VAD-based Community Apps

- Example of VAD ONNX Runtime model usage in [C++](#)
- Example of VAD [ExecuTorch](#) model usage in [C++](#)
- Voice activity detection for the [browser](#) using ONNX Runtime Web
- [Rust](#), [Rust \(wavekat-vad\)](#), [Go](#), [Java](#), [C++](#), [C#](#) and [other](#) community examples
- OpenVINO conversion [guidelines](#)
- A tinygrad [model](#) with a pico example in the docsting + separate [weights](#) in safetensors format (for simplicity we provided just the 16k model)